

11. Assume that the reference variable `r` refers to a serializable object. Write code that serializes the object to the file `ObjectData.dat`.

### Short Answer

1. What is meant when it is said that an exception is thrown?
2. What does it mean to catch an exception?
3. What happens when an exception is thrown, but the `try` statement does not have a catch clause that is capable of catching it?
4. What is the purpose of a `finally` clause?
5. Where does execution resume after an exception has been thrown and caught?
6. When multiple exceptions are caught in the same `try` statement and some of them are related through inheritance, does the order in which they are listed matter?
7. What types of objects can be thrown?
8. When are you required to have a `throws` clause in a method header?
9. What is the difference between a checked exception and an unchecked exception?
10. What is the difference between the `throw` statement and the `throws` clause?
11. What is the difference between a text file and a binary file?
12. What is the difference between a sequential access file and a random access file?
13. What happens when you serialize an object? What happens when you deserialize an object?

## Programming Challenges

MyProgrammingLab™ Visit [www.myprogramminglab.com](http://www.myprogramminglab.com) to complete many of these Programming Challenges online and get instant feedback.

### 1. TestScores Class

Write a class named `TestScores`. The class constructor should accept an array of test scores as its argument. The class should have a method that returns the average of the test scores. If any test score in the array is negative or greater than 100, the class should throw an `IllegalArgumentException`. Demonstrate the class in a program.

### 2. TestScores Class Custom Exception

Write an exception class named `InvalidTestScore`. Modify the `TestScores` class you wrote in Programming Challenge 1 so that it throws an `InvalidTestScore` exception if any of the test scores in the array are invalid.

### 3. RetailItem Exceptions

Programming Challenge 4 of Chapter 6 required you to write a `RetailItem` class that holds data pertaining to a retail item. Write an exception class that can be instantiated and thrown when a negative number is given for the price. Write another exception class that can be instantiated and thrown when a negative number is given for the units on hand. Demonstrate the exception classes in a program.

#### 4. Month Class Exceptions

Programming Challenge 5 of Chapter 8 required you to write a `Month` class that holds information about the month. Write exception classes for the following error conditions:

- A number less than 1 or greater than 12 is given for the month number.
- An invalid string is given for the name of the month.

Modify the `Month` class so that it throws the appropriate exception when either of these errors occurs. Demonstrate the classes in a program.

#### 5. Payroll Class Exceptions

Programming Challenge 5 of Chapter 6 required you to write a `Payroll` class that calculates an employee's payroll. Write exception classes for the following error conditions:

- An empty string is given for the employee's name.
- An invalid value is given for the employee's ID number. If you implemented this field as a string, then an empty string would be invalid. If you implemented this field as a numeric variable, then a negative number or zero would be invalid.
- An invalid number is given for the number of hours worked. This would be a negative number or a number greater than 84.
- An invalid number is given for the hourly pay rate. This would be a negative number or a number greater than 25.

Modify the `Payroll` class so that it throws the appropriate exception when any of these errors occurs. Demonstrate the exception classes in a program.

#### 6. FileArray Class

Design a class that has a static method named `writeArray`. The method should take two arguments: the name of a file and a reference to an `int` array. The file should be opened as a binary file, the contents of the array should be written to the file, and then the file should be closed.

Write a second method in the class named `readArray`. The method should take two arguments: the name of a file and a reference to an `int` array. The file should be opened, data should be read from the file and stored in the array, and then the file should be closed. Demonstrate both methods in a program.

#### 7. File Encryption Filter

File encryption is the science of writing the contents of a file in a secret code. Your encryption program should work like a filter, reading the contents of one file, modifying the data into a code, and then writing the coded contents out to a second file. The second file will be a version of the first file, but written in a secret code.

Although there are complex encryption techniques, you should come up with a simple one of your own. For example, you could read the first file one character at a time, and add 10 to the character code of each character before it is written to the second file.

#### 8. File Decryption Filter

Write a program that decrypts the file produced by the program in Programming Challenge 7. The decryption program should read the contents of the coded file, restore the data to its original state, and write it to another file.

### 9. TestScores Modification for Serialization

Modify the `TestScores` class that you created for Programming Challenge 1 to be serializable. Write a program that creates an array of at least five `TestScore` objects and serializes them. Write another program that deserializes the objects from the file.

### 10. Exception Project



VideoNote

The Exception  
Project  
Problem

This assignment assumes you have completed Programming Challenge 1 of Chapter 10 (`Employee` and `ProductionWorker` Classes). Modify the `Employee` and `ProductionWorker` classes so they throw exceptions when the following errors occur:

- The `Employee` class should throw an exception named `InvalidEmployeeNumber` when it receives an invalid employee number.
- The `ProductionWorker` class should throw an exception named `InvalidShift` when it receives an invalid shift.
- The `ProductionWorker` class should throw an exception named `InvalidPayRate` when it receives a negative number for the hourly pay rate.

Write a test program that demonstrates how each of these exception conditions works.